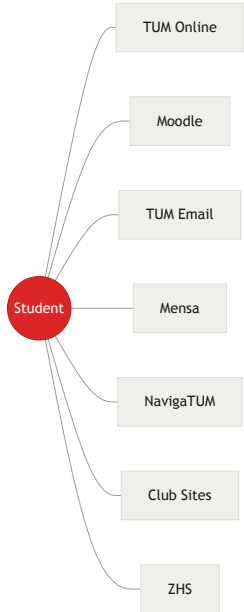
AssisTUM

Your Autonomous Campus Co-Pilot

REPLY Makeathon 2026 — Team Pui

Max Solovyev, Nicolas Mallinckrodt, Kho Yin Xun

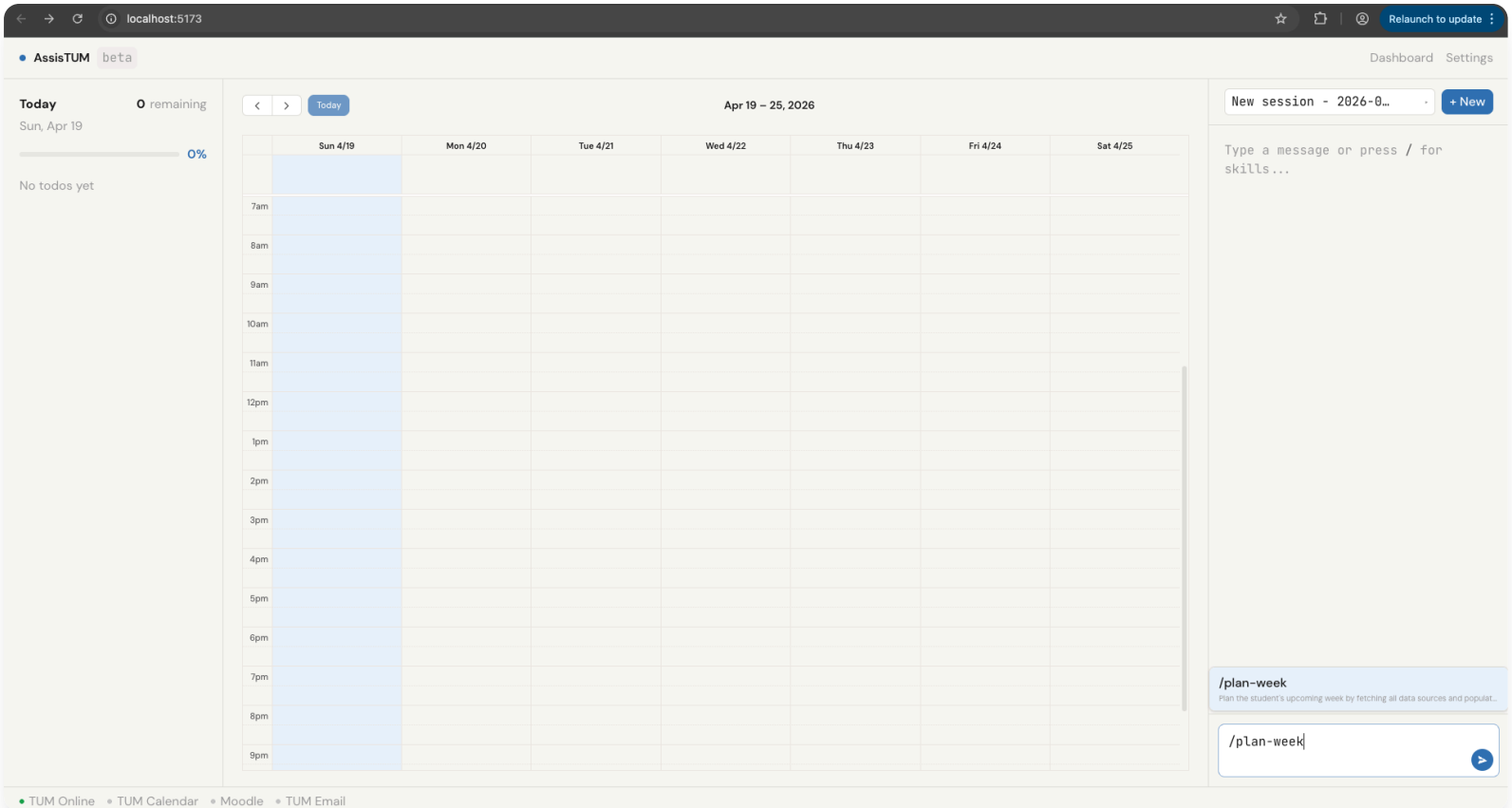
Students are **human APIs**



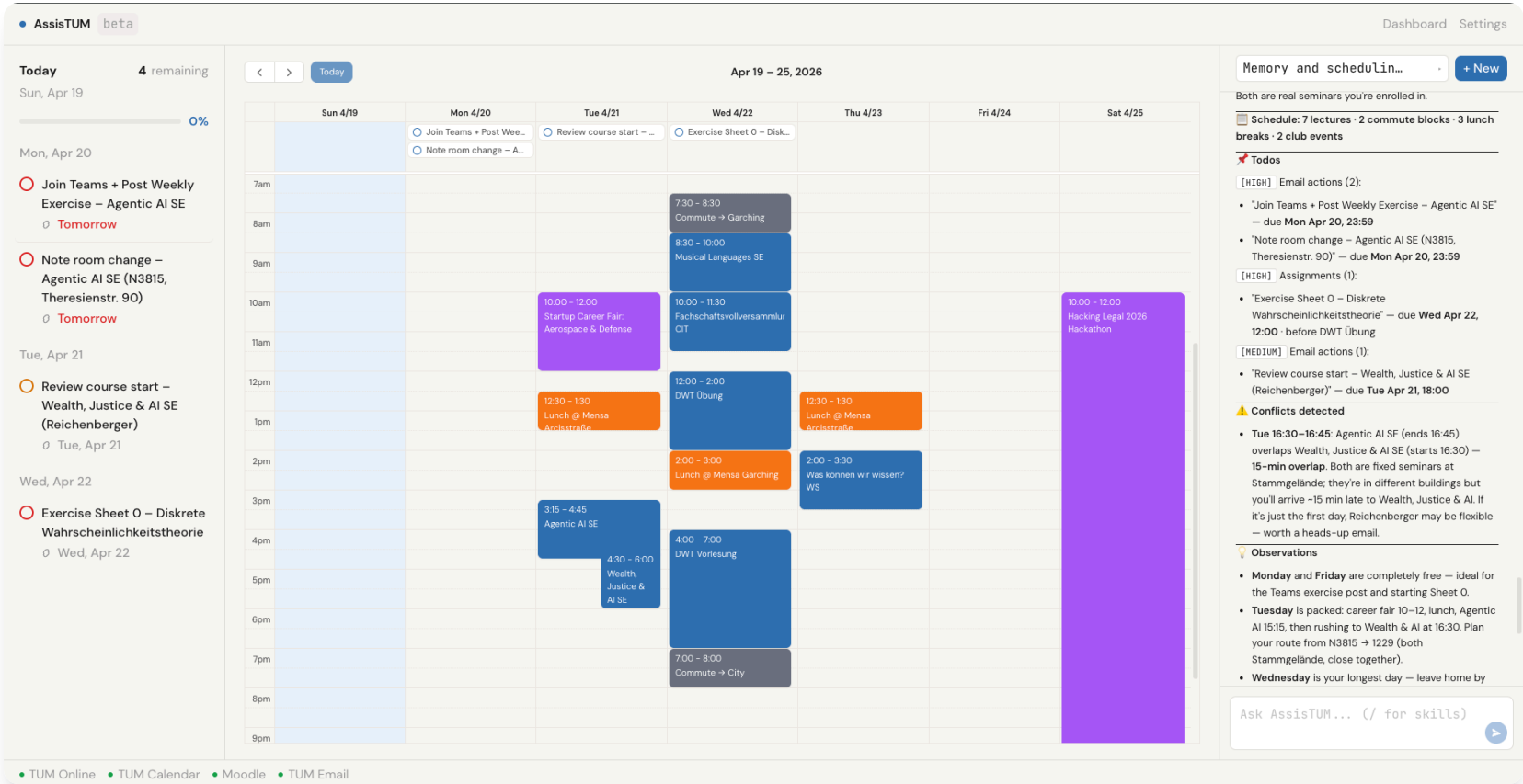
30+ minutes every week

Systems don't talk to each other. Students are the glue.

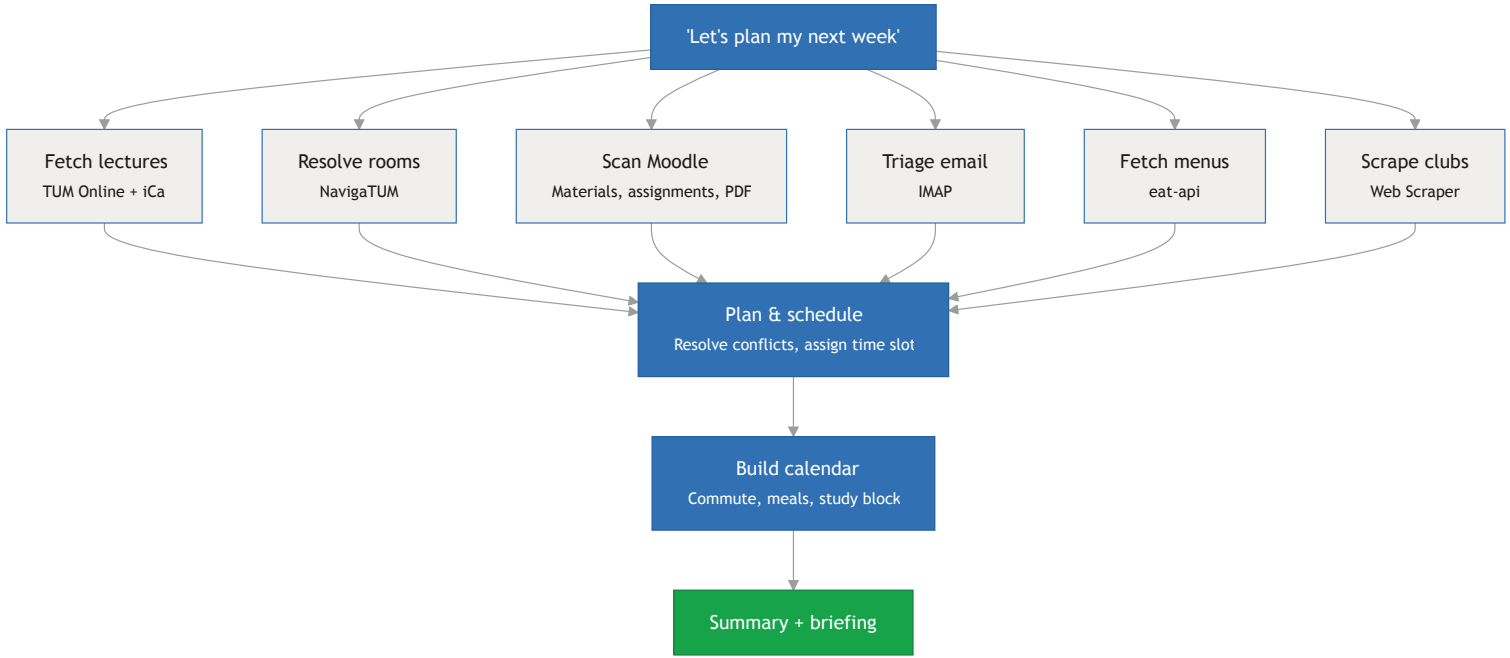
What if it took **one message**?



3 minutes later



One message. **Seven autonomous phases.**



Each phase hits a **real external system**. No mocks. No pre-seeded data.

It doesn’t just fetch data — it **makes decisions**

LECTURES (CALENDAR)

Imported real schedule from TUM Online + iCal, placed every course on the calendar. **Automatic weekly setup.**

COURSE MATERIALS (MOODLE)

Fetched materials, assignments, and PDFs — extracted text, summarized, linked in tasks. **Reads and understands content.**

EMAIL

Read university inbox, identified actionable items, created tasks with deadlines. **Autonomous triage.**

MEMORY (COGNEE)

Remembers preferences, past choices, and context across sessions. **Learns and adapts over time.**

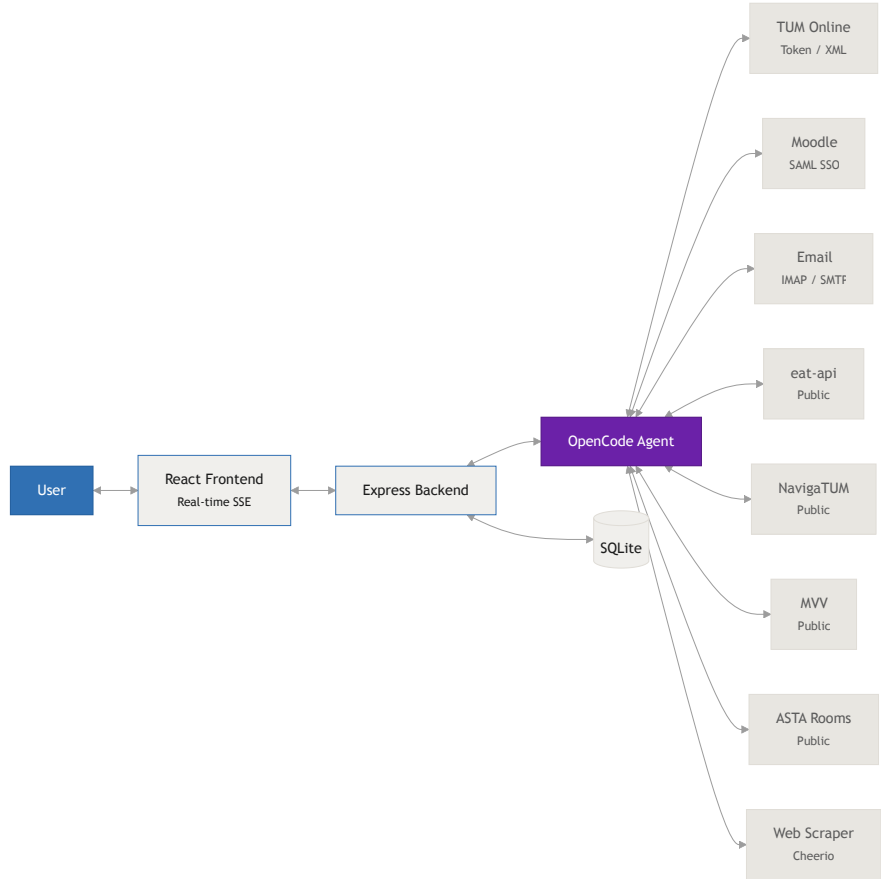
COMMUTE

Looked up every room in NavigaTUM, determined campus, added travel time. **No one told it to.**

MENSA

Picked the closest canteen to your location that day. Chose a meal. **Context-aware scheduling.**

MCP-Powered Agent Architecture



8 University Systems. **Real APIs. Real Auth.**

System	Auth	Autonomous Actions
Moodle	SAML SSO	Fetches assignments, **downloads PDFs, extracts text, summarizes**
TUM Online	Token	Pulls lectures, syncs courses, fetches grades
NavigaTUM	Public	Resolves room codes → campus, **auto-generates commute blocks**
Email	IMAP/SMTP	Reads inbox, **triages into actionable tasks with deadlines**
Mensa	Public	Fetches menus, **picks closest canteen by schedule context**
MVV	Public	Live departures, **calculates when to leave**
Clubs	Web scrape	Extracts events from **arbitrary club websites**
Study Rooms	ASTA API	**Real-time availability** across campuses

30 minutes → **3 minutes**

Before

- 6 browser tabs open
- Manual copy-paste between systems
- Missed deadlines
- Forgotten lunches
- No travel time planning

After

- One conversation
- Full week planned
- Every deadline tracked
- Campus-aware meals
- Commute blocks included

Works with **any TUM student's real credentials**, today. Not a prototype — a **working product**.

Not a chatbot. A **campus operating system.**

DATA INTELLIGENCE

- Fetches and reads PDFs, exercise sheets, and lecture slides from Moodle
- Extracts text from PDFs, summarizes content, links in tasks
- Scrapes any student club website for events
- Reads and triages university email

REAL-TIME INFORMATION

- Live MVV departures — when to leave and which train
- Study room availability across campuses
- Mensa occupancy — how crowded right now?
- Canteen menus for the full week

AUTONOMOUS ACTIONS

- Resolves lecture rooms → determines campus → adds commute time
- Picks closest mensa by your schedule, selects meals
- Schedules study sessions around your deadlines
- Detects and resolves calendar conflicts

AGENT SKILLS

- `/plan-week` — Build full week from 5+ systems
- `/review-lectures` — Review sessions with Moodle materials
- `/schedule-study-sessions` — Auto-schedule before deadlines
- `/commute-helper` `/find-study-room` `/conflict-resolver` `/course-brainstorm`

Limitations & Honest Tradeoffs

LLM Hallucinations

The agent can misinterpret data or generate incorrect summaries. Every action it takes — events, todos, meal picks — should be **reviewed by the student**, not blindly trusted.

Omissions

The agent may miss an assignment, skip a club event, or overlook a deadline. It's a **co-pilot, not autopilot** — it reduces work, but doesn't replace human judgment.

API Fragility

University systems change without notice. SAML sessions expire, Moodle restructures pages, club websites update layouts. The agent handles failures gracefully, but **no integration is guaranteed forever**.

We built AssisTUM to **assist**, not to replace. The student always has the final say.

AssisTUM

Being a student should be about learning — not logistics.

Built in <40 hours. Ready to deploy.

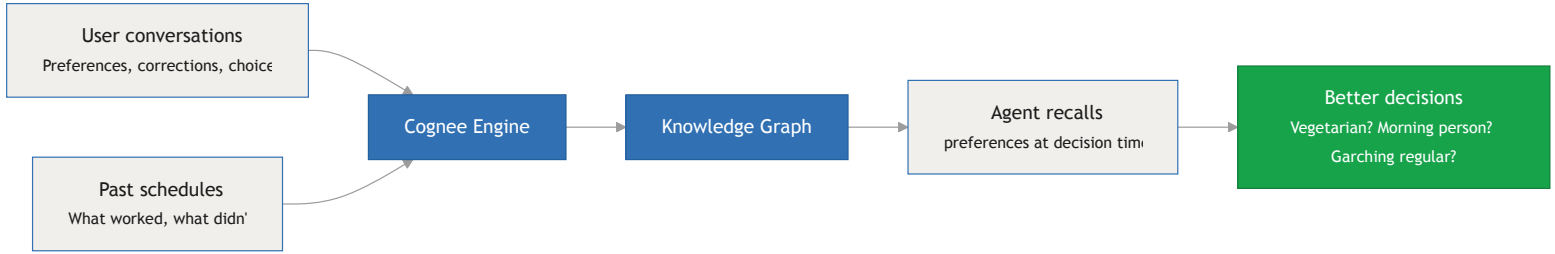
Team Pui — REPLY Makeathon 2026

Appendix

Deep-dive into each integration

A1

Memory — Cognee Integration



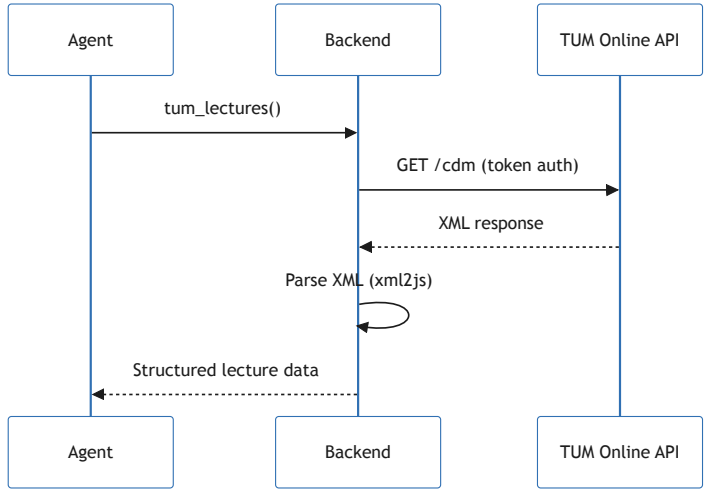
How it works: As the student interacts with AssisTUM, Cognee extracts preferences and patterns from conversations and stores them in a knowledge graph. The agent queries this graph at decision time.

Examples: "Prefers vegetarian meals" · "Usually studies in the morning" · "Takes U6 not U3" · "Skips Mensa on Fridays"

Result: The agent gets smarter over time — no explicit settings, just natural conversation.

A2

TUM Online Integration

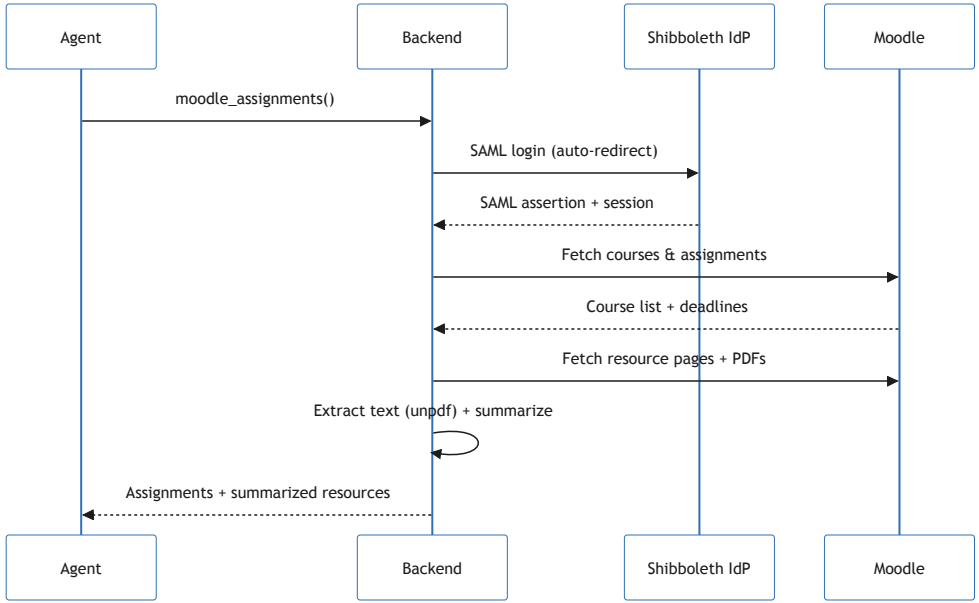


Auth: Token-based (email confirmation flow) | **Data:** XML → xml2js → JSON → calendar events

Capabilities: Fetch lecture schedule, sync courses to local DB, fetch grades

A3

Moodle Integration



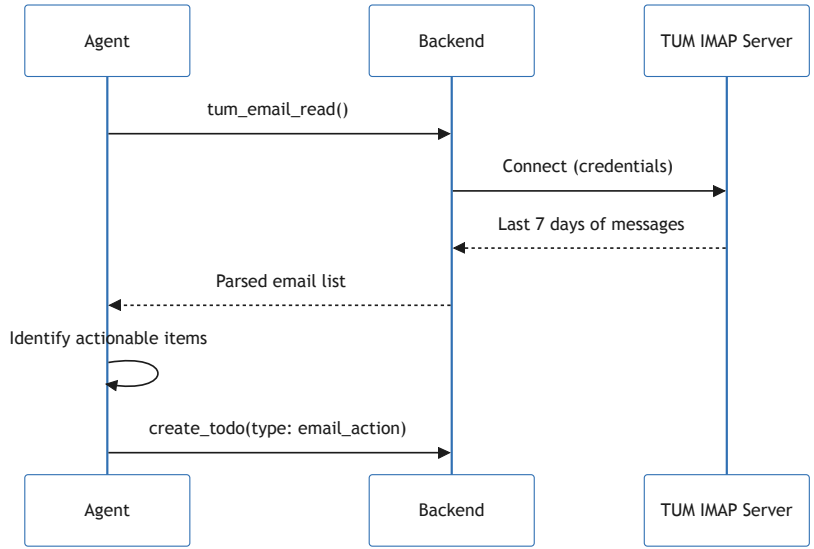
Auth: SAML Shibboleth SSO — auto-redirect chain, session caching, auto-refresh on expiry

PDF Pipeline: Moodle page → detect PDF → download → extract text → summarize → link in task

Most technically complex integration. SAML auth alone is non-trivial.

A4

Email Integration



Auth: IMAP credentials + SMTP for sending | **Read:** Last N days, extract sender/subject/date/body

Agent behavior: Scans for actionable items, creates todos with 48h default deadline unless email specifies one

A5

Mensa & Canteen

API: TUM eat-api (public, no auth)

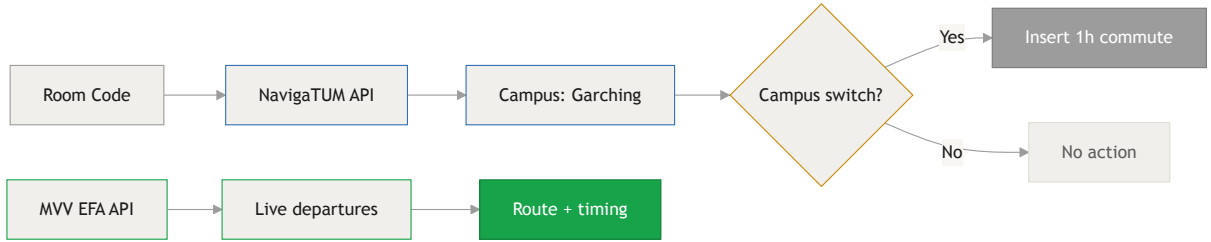
Capabilities:

- Weekly menus by location (mensa-garching, mensa-arcisstr, etc.)
- Live occupancy head count

Agent behavior: Cross-references calendar → determines campus → picks closest canteen → schedules lunch → selects meal

A6

NavigaTUM & Commute



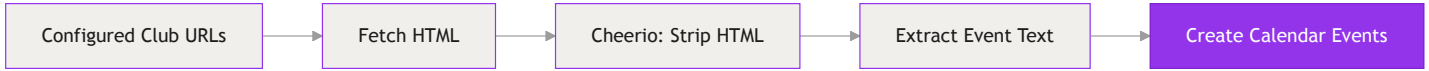
APIs: NavigaTUM (room/building search) + MVV EFA (real-time departures)

Commute logic: For each lecture → resolve room code → determine campus → if campus switch, insert 1h commute block

Route planning: NavigaTUM room lookup + MVV live departures → tells student which U-Bahn to take and when to leave

A7

Student Clubs



Method: Configurable club URLs + Cheerio HTML scraping

Capabilities: Scrapes arbitrary club websites, strips HTML, extracts event text

Resilience: Handles errors gracefully — skips clubs that fail, continues with rest

Works with **any** student club website — no special API needed

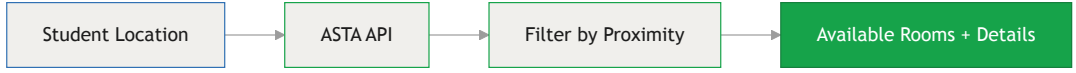
A8

Study Rooms

API: ASTA study room API (public)

Capabilities:

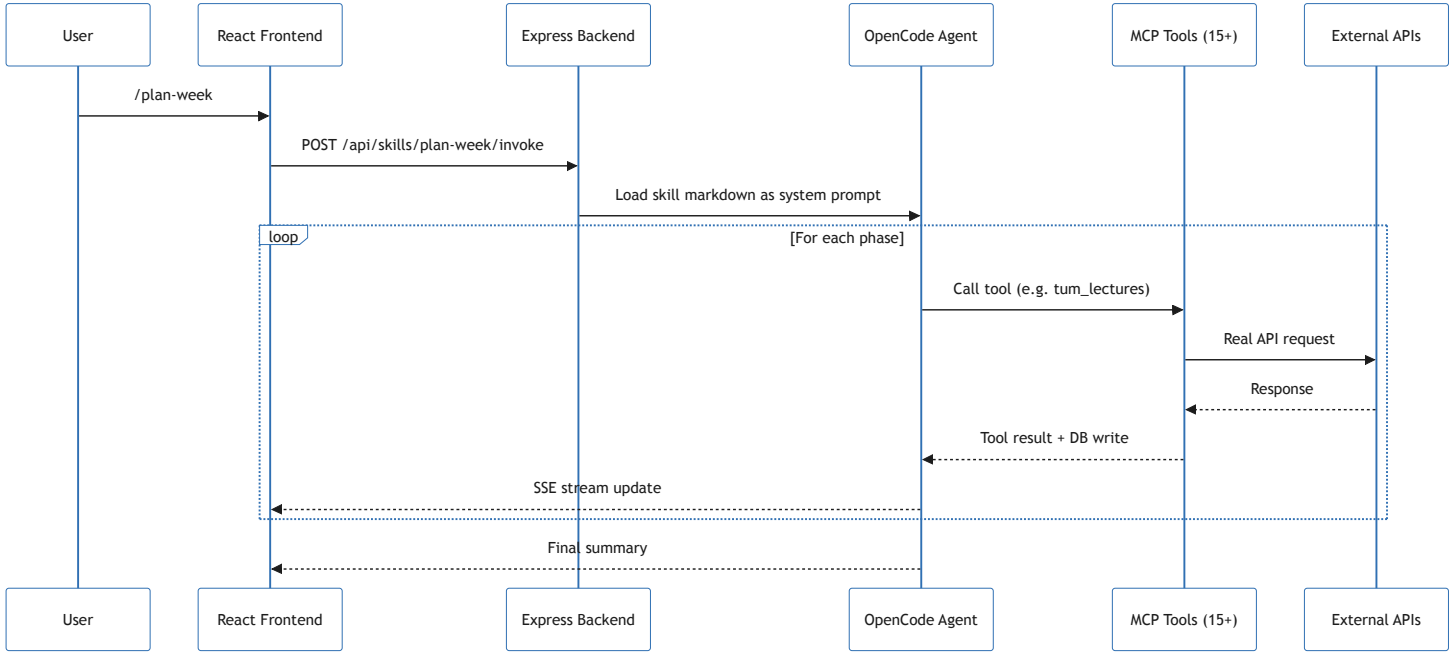
- Real-time room availability across TUM campuses
- Filters by proximity to student's current/upcoming location
- Reports available rooms with building details



Accessed via `/find-study-room` slash command or natural language request

A9

The Agent Engine

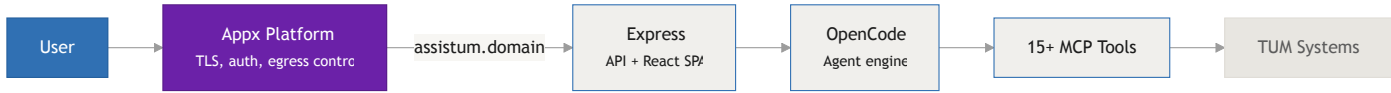


Engine: OpenCode SDK | **Protocol:** MCP — 15+ tools | **Skills:** 7 markdown workflows, loaded dynamically

Error handling: Per-tool graceful degradation — if one integration fails, agent skips and continues

A10

Hosting on Appx



Appx is a self-hostable platform for building, sharing, and running agentic applications.

AssisTUM deploys to Appx as a managed project — zero infra work:

- **TLS + Auth** — Appx handles HTTPS and platform login
- **Subdomain routing** — `assistum.<your-domain>` out of the box
- **Egress control** — monitors and restricts agent outbound traffic
- **One command** — `task start:appx` launches everything

AssisTUM is the first agentic app running on Appx — proving that complex, multi-system agents can be packaged and shared as easily as traditional web apps.